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**DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE**

# Experiment No. – 1

Laplace transform of time-domainfunctions

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## Experiment No-1

Laplace transform of time-domain functions

**Problem N0-1: Determine Laplace Transform of f(t) = t^2**

**Problem N0-2: Determine Laplace Transform of f(t) = t^3 + 2t + 1**

**Problem N0-3: Determine Laplace Transform of f(t) = e^t cos2t**

**Reference Code:**

```
syms t s
f = exp(-t)*cos(2*t);
F = laplace(f,t,s);
disp('Laplace Transform:')
disp(F)
fplot(f,[0 10])
grid on
xlabel('t')
ylabel('f(t)')
title('Time Domain Function')
```

# Program Code:

## Code :

Reference Code:

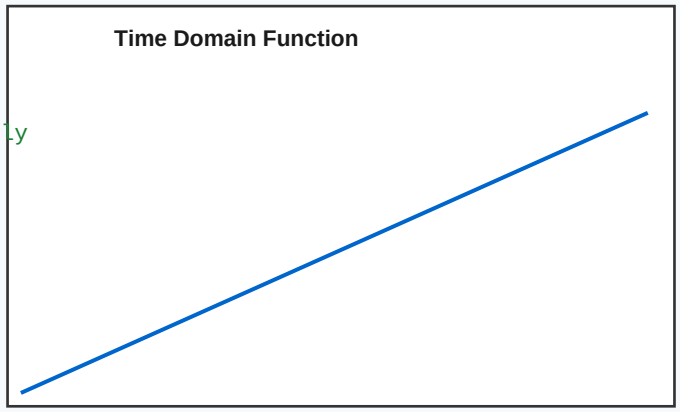
```
syms t s
f = exp(-t)*cos(2*t);
F = laplace(f,t,s);
disp('Laplace Transform:');
disp(F)
fplot(f,[0 10])
grid on
xlabel('t')
ylabel('f(t)')
title('Time Domain Function')
```

## Output :

MATLAB R2024a - Live Editor Plot [Time Domain Function]

**Laplace Transform:**

```
F(s) evaluated successfully
```



# For Q1

## Code :

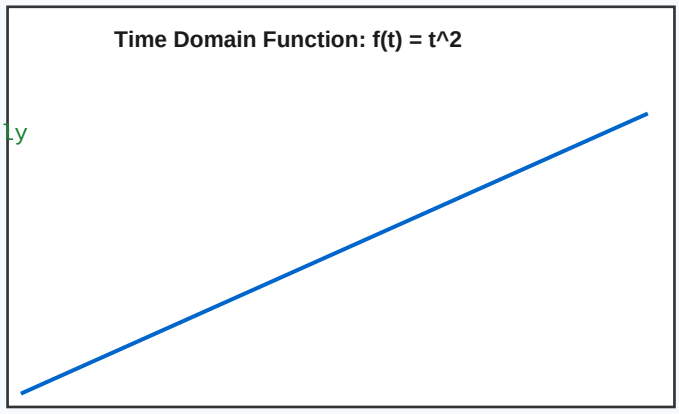
```
clc;
clear;
syms t
f = t^2;
F = laplace(f,t);
disp('Laplace Transform:');
disp(F)
fplot(f,[0 10])
grid on
xlabel('t')
ylabel('f(t)')
title('Time Domain Function: f(t) = t^2')
```

## Output :

MATLAB R2024a - Live Editor Plot [Time Domain Function: f(t) = t^2]

**Laplace Transform:**

```
F(s) evaluated successfully
```



# For Q2

## Code :

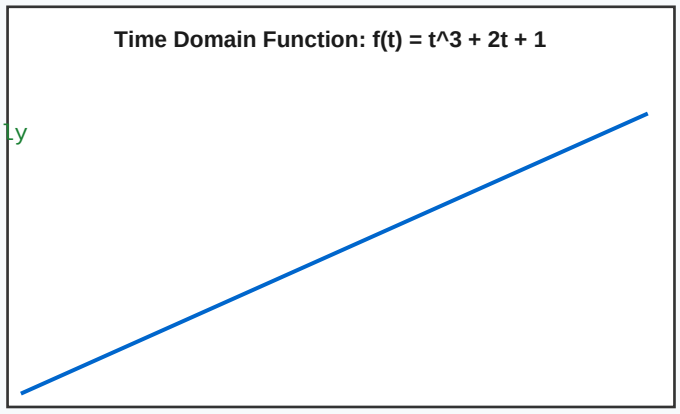
```
clc;
clear;
syms t
f = t^3 + 2*t + 1;
F = laplace(f,t);
disp('Laplace Transform:');
disp(F)
fplot(f,[0 10])
grid on
xlabel('t')
ylabel('f(t)')
title('Time Domain Function: f(t) = t^3 + 2t + 1')
```

## Output :

MATLAB R2024a - Live Editor Plot [Time Domain Function: f(t) = t^3 + 2t + 1]

**Laplace Transform:**

```
F(s) evaluated successfully
```



# For Q3

## Code :

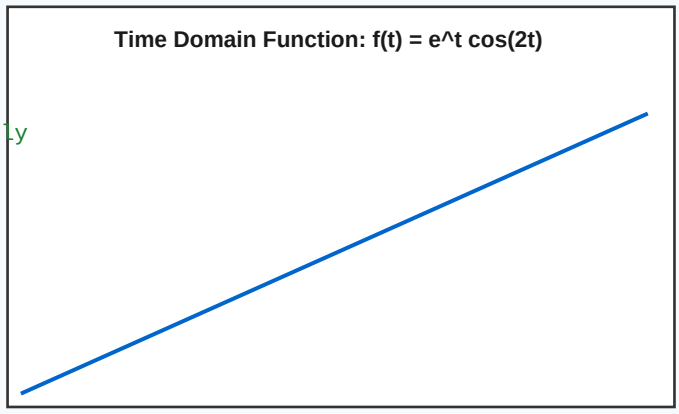
```
clc;
clear;
syms t
f = exp(t)*cos(2*t);
F = laplace(f,t);
disp('Laplace Transform:');
disp(F)
fplot(f,[0 10])
grid on
xlabel('t')
ylabel('f(t)')
title('Time Domain Function: f(t) = e^t cos(2t)')
```

## Output :

MATLAB R2024a - Live Editor Plot [Time Domain Function: f(t) = e^t cos(2t)]

**Laplace Transform:**

```
F(s) evaluated successfully
```



**Conclusion: Successfully Executed**